Evaluate yourself A solution of a salt of metal was electrolysed for 15 minutes with a current of 0.15 amperes. The mass of the metal deposited at the cathode is 0.783g. calculate the equivalent mass of the metal.

Batteries

Batteries are indispensable in the modern electronic world. For example, Li – ion batteries are used in cell phones, dry cell in flashlight etc…. These batteries are used as a source of direct current at a constant voltage. We can classify them into primary batteries (non – rechargeable) and secondary batteries (rechargeable). In this section, we will briefly discuss the electrochemistry of some batteries.

Leclanche cell

Anode : Zinc container

Cathode : Graphite rod in contact with MnO_2

Electrolyte : ammonium chloride and zinc chloride in water

Emf of the cell is about 1.5V

Cell reaction

Oxidation at anode

$$Zn\,(s) \rightarrow Zn^{2+}(aq) + 2e^- \qquad .....(1)$$

Reduction at cathode

$$2\,NH_4^+(aq) + 2e^- \rightarrow 2NH_3\,(aq) + H_2(g) \qquad .....(2)$$

The hydrogen gas is oxidised to water by MnO_2

$$H_2\,(g) + 2\,MnO_2\,(s) \rightarrow Mn_2O_3\,(s) + H_2O\,(l) \qquad .....(3)$$

Equation (1) + (2)+(3) gives the overall redox reaction

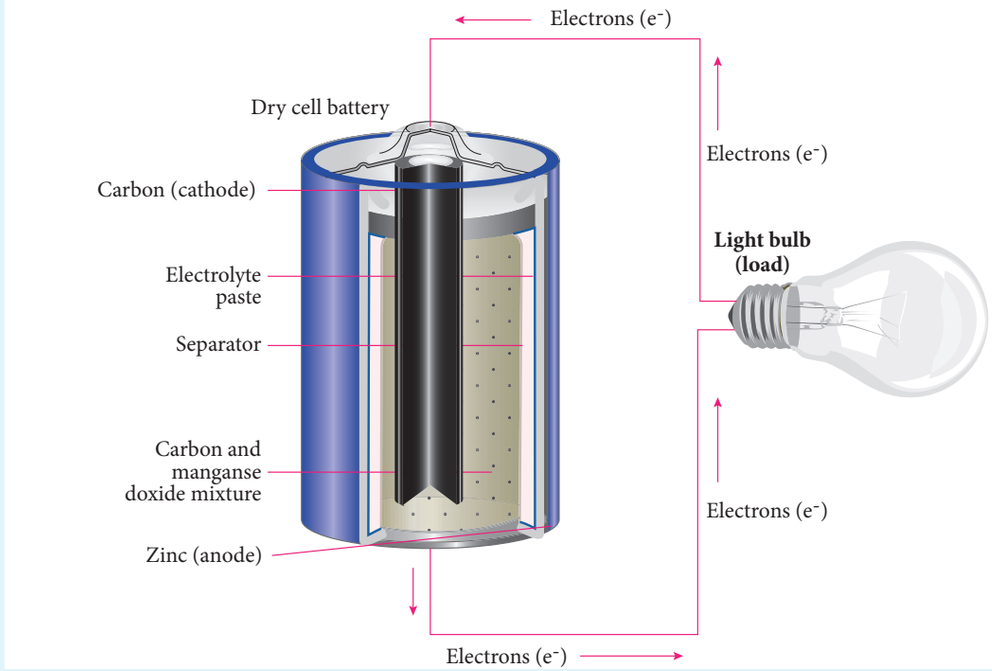


Figure 9.10 Leclanche cell

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